

Problem-1: Derivation of Rate Expression from the Reaction Mechanism

A) Nitrogen Pentoxide Decomposition

Consider the decomposition of nitrogen pentoxide into nitrogen dioxide and oxygen,



This is not an elementary reaction and its mechanism consists of the following set of elementary reactions.



The rate constants for the reactions R1a, R1b, R1c and R1d are denoted as k_1 , k_2 , k_3 and k_4 , respectively. Here, NO and NO_3 are regarded as the intermediate species.

(a) Derive the rate expression for the overall reaction R1 from its mechanism (R1a-d) using the pseudo- steady state approximation.

(b) What is the overall order of the reaction?

B) Hydrogen Bromide Formation

Consider the formation of hydrogen bromide given by,



It is not an elementary reaction and it's mechanism can be written as



The rate constants for the reactions R2a, R2b, R2c, R2d and R2e are denoted as k_1 , k_2 , k_3 , k_4 and k_5 , respectively. Derive the rate expression for R2 from its mechanism using the pseudo-steady state approximation. (10 points)